

Simulated Irregular Longitudinal Data

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This script generates irregular longitudinal measurements together with a subject-level binary outcome.

Subjects

For subjects

$$i = 1, \dots, n,$$

the simulation uses

$$n = 10.$$

Each subject has a random number of repeated measurements,

$$m_i \sim \text{DiscreteUniform}\{3, \dots, 10\}.$$

Observation Times

Conditional on m_i , observation times are independently generated as

$$t_{ij} \sim \text{Uniform}(0, 10), \quad j = 1, \dots, m_i,$$

and then sorted within subject.

Thus, subjects can have different numbers of observations and measurements occur at irregular times.

Subject-Level Latent Variable

Each subject has a latent variable

$$z_i \sim N(0, 1).$$

The same z_i affects both the longitudinal trajectory and the binary outcome.

Longitudinal Process

The longitudinal measurement for subject i at observation j is

$$X_{ij} = \beta_0 + \beta_1 t_{ij} + \beta_2 z_i + \epsilon_{ij},$$

where

$$\epsilon_{ij} \sim N(0, \sigma^2).$$

The parameter values are

$$\beta_0 = 1, \quad \beta_1 = 0.5, \quad \beta_2 = 1.5, \quad \sigma = 0.5.$$

Therefore,

$$X_{ij} = 1 + 0.5t_{ij} + 1.5z_i + \epsilon_{ij}, \quad \epsilon_{ij} \sim N(0, 0.25).$$

Binary Outcome

Each subject has a single binary outcome

$$Y_i \sim \text{Bernoulli}(p_i),$$

with

$$\text{logit}(p_i) = \alpha_0 + \alpha_1 z_i,$$

where

$$\alpha_0 = -0.5, \quad \alpha_1 = 1.$$

Hence,

$$p_i = \frac{\exp(-0.5 + z_i)}{1 + \exp(-0.5 + z_i)}.$$

Because z_i appears in both X_{ij} and Y_i , the longitudinal trajectory and binary outcome are associated through the shared subject-level latent variable.

Output Files

The script saves:

- 'longitudinal_data.csv': repeated-measurement

data with columns 'subject', 'time', 'x', and 'z'.
- 'subject_data.csv': one row per subject with
columns 'subject', 'z', 'event', and 'event_probability'.

Ex. 1 1.

Ex. 2

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Ex. 3